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Additive Manufacturing for Space and Aerospace Applications (Prof. Tommaso Ghidini)



Optimization of the support bracket for the payload adapter of Vega launcher
TEAM 01

Ben Dahou Yassir [REDACTED]
Henriquet Diego [REDACTED]
Pratola Oscar [REDACTED]
Spinelli Giulia Leda [REDACTED]
Verri Matteo [REDACTED]

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Nomenclature

$\hat{x}$	x-axis Unit Vector
$\hat{y}$	y-axis Unit Vector
ω	Natural Frequency $[Hz]$
m	mass $[kg]$
QSF	Qualification Safety Factor
SF	Safety Factor

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Chapter - 0

Introduction

0.1 Component Overview

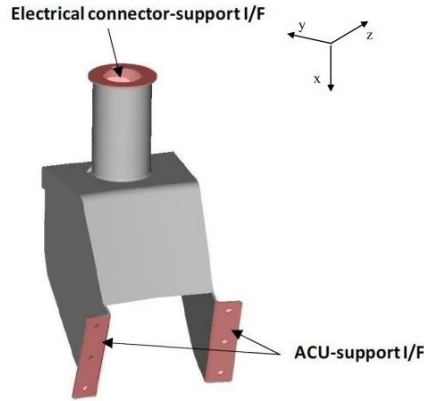


Figure 0.1: Electrical Connector Support

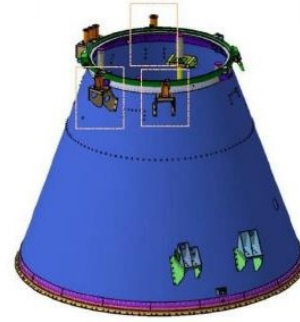


Figure 0.2: Upper Stage Adapter

The Vega launcher comes with an off-the-shelf adapter to connect the spacecraft to the launcher's upper stage. The adapter is a conical shell with interfaces on both bases and supports on the side to connect the umbilicals [1]. The support itself is originally composed of four bolted pieces. In this redesign, the goal is to develop a much lighter single component capable to fulfill the constraints presented in Section 0.2. The interfaces with the electrical connector on one side and the shell of the adapter on the other side, must be preserved and are highlighted in *red* in Figure 0.1.

0.2 Loads and Constraints

The support is subjected to the following loads:

Type	Direction	Value
acceleration	$\hat{x}$	15 <i>g</i>
acceleration	$\hat{y}$	$\pm 10g$
point mass	[-]	0.300 <i>kg</i>

Table 1: Loads Overview

The constraints on natural frequencies are set as follows:

	Value	Unit
$\omega_{\hat{x}}$	> 100	Hz
$\omega_{\hat{y}}$	> 50	Hz

Table 2: Constraints on Natural Frequency ω

0.3 Printing Material

The printing material selected is AlSi10Mg an aluminum alloy known for its lightweight and strong characteristics; the corrosion resistance ensures that the adapter can withstand the harsh environmental conditions experienced during launch. It has good machinability and high compatibility with other aluminum alloys or composite materials often used in launchers. Furthermore, its high strength-to-weight ratio allows for the construction of robust yet lightweight components, which is essential for space launch systems.

Parameter	Value	Unit	Parameter	Value	Unit
Young Modulus	70.000	GPa	Density	2.680	$\frac{g}{cm^3}$
Poisson's Coefficient	0.33	[-]	Yield Stress	220.000	MPa
Shear Strength	26.300	GPa			

Table 3: Material Properties

Step - 1

Design for performance

The optimization process aims to reduce the overall component mass, satisfying the acceleration and natural frequency constraints, shown in Table 1, by performing a topology optimization. The process is carried out with the software Autodesk Fusion using the "Generative Design" feature. This software is preferred instead of Altair Inspire or nTop, since their topology optimization is less efficient in terms of mass reduction and computationally because keeps overhangs non-attached to the connector leaving them mid-air. Before it begins, selected regions of the component must remain untouched during the process because they are either connected to the upper stage adapter (fastening connectors) or are the electrical connector support (upper cylinder). These sections constitute the frozen domain of the design. For the six portions of the domain where the component is fixed to the launcher adapter, a thickness of $3mm$ is imposed for each interface. Instead, for the connector interface, the whole upper cylinder coming out from the upper part of the design domain is imposed as the second frozen domain. In addition, it is decided to add obstacle regions where the bolts and the connector are placed to ensure a good interface for these components with the bracket. Consequently, the remaining component volume is selected as the design domain, where the topology optimization is applied.

Fixed constraints are applied to the six bolts and the point mass of 300 g is fixed to the electrical connector. Regarding the acceleration loads mentioned in Table 1, they are increased by a qualification safety factor of $QSF = 1.25$, to be compliant with the Vega User's Manual[1]. Moreover, for each load case, a combination between the acceleration load along the $x - axis$ and one of the two along the $y - axis$ has been considered.

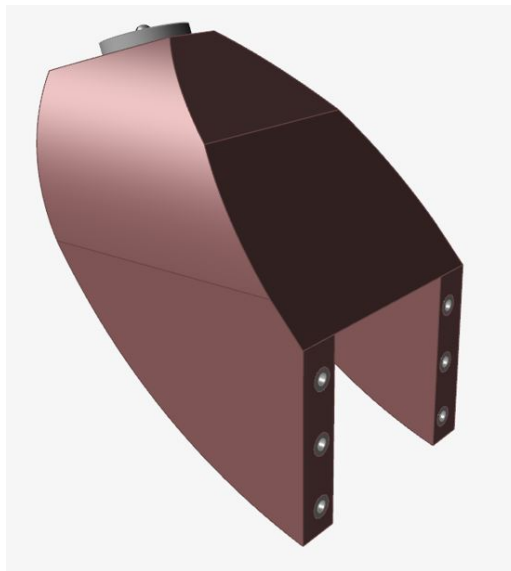


Figure 1.1: Design domain (maroon) and frozen domain (gray)

At this point two possible analyses can be performed, a mass minimization analysis and a stiffness maximization analysis. Both of them are carried out to understand which one can lead to the best result in the least number of cycles. For both analyses, a constraint on the primary frequency mode has been added to keep it above 100 Hz , in order to respect all the requirements. In addition, for the mass minimization problem, the objective has been set as a safety factor of $SF = 1$. Instead, for the stiffness maximization problem the objective is to reach a mass of $m = 100\text{ g}$ with a safety factor of $SF = 1$.

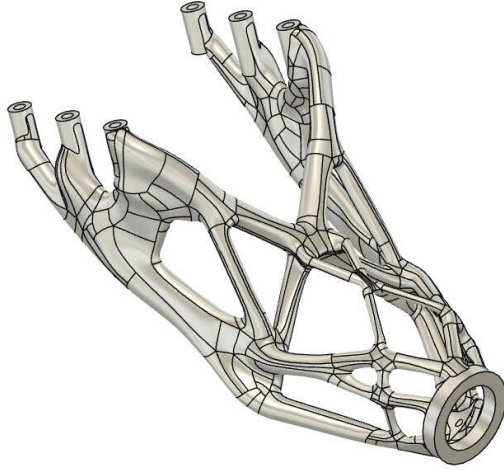


Figure 1.2: Mass minimization result after the first cycle

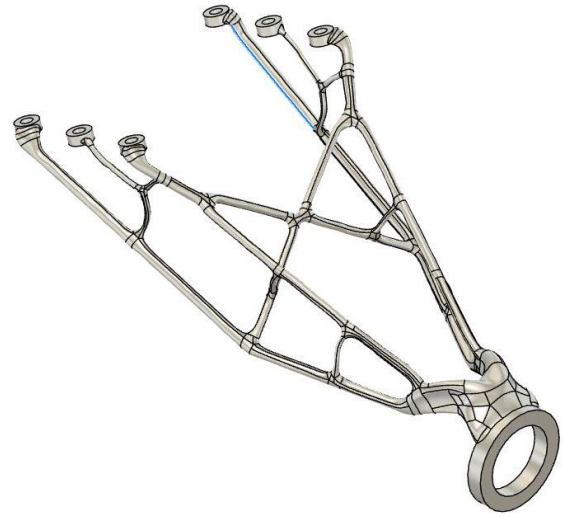


Figure 1.3: Stiffness maximization result after the first cycle

The results reported in Figures 1.2 and 1.3 show how the stiffness maximization problem is more proficient at removing mass considering the same safety factor. Therefore, the stiffness maximization problem is selected for the following optimization analyses. Starting from the result presented in Figure 1.3, two more refinement cycles are performed, using as starting geometry the results coming from the previous analysis, preserving all the constraints and the loads.

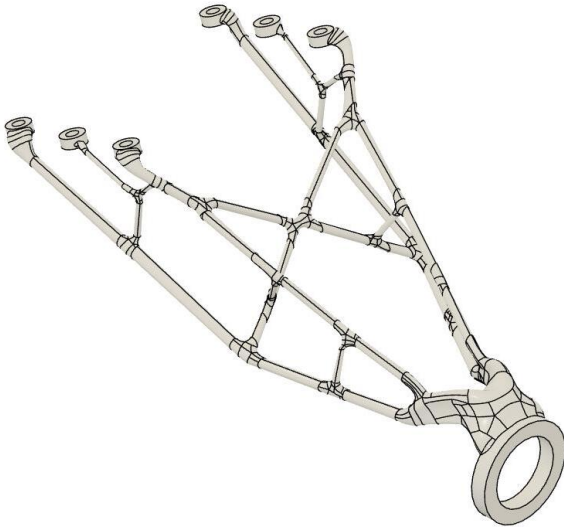


Figure 1.4: Stiffness maximization result after the third cycle

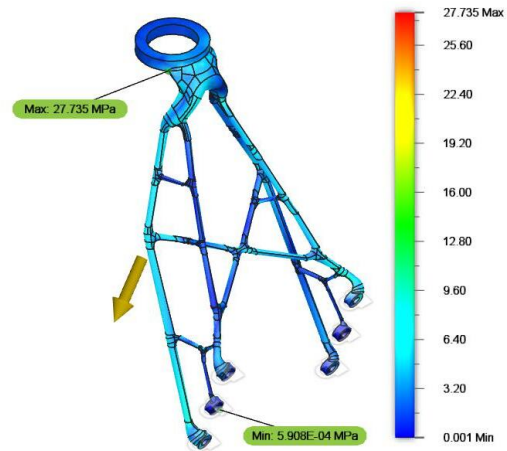


Figure 1.5: Static simulation for the third cycle of the stiffness maximization analysis

Figure 1.4 shows the result obtained at the third cycle of refinement. At this point, the bracket's weight is $m = 109.387 \text{ g}$. One thing that can be noticed is that the bolts in the middle are connected to the body with very thin beams. This can be interpreted as the fact that the contribution of these two bolts to the load carrying of the structure is negligible compared to the other four bolts. Moreover, from Figure 1.5 it can be seen that the maximum internal stress is 27.735 MPa , far below the critical one for the material considered in Table 3. For the reasons mentioned above, it is decided to start the optimization process again, removing the passive region constraint for the middle bolts, to see if it is possible to reduce the mass even more.

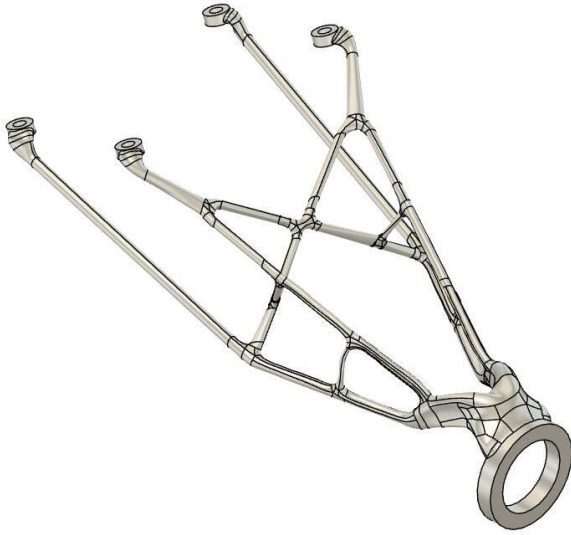


Figure 1.6: Stiffness maximization result after the first cycle for the four bolts configuration

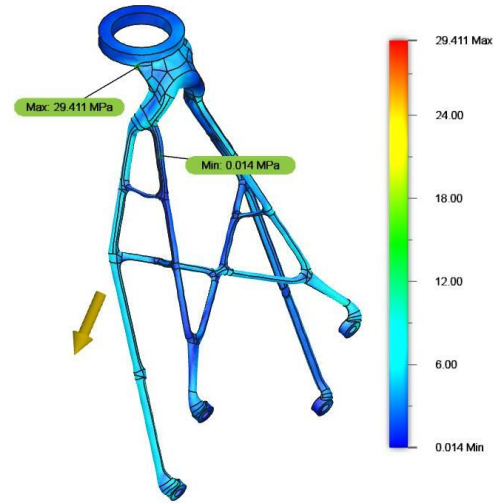


Figure 1.7: Static simulation for the first cycle of the four bolts configuration

As it is possible to see in Figure 1.6 the result is very similar to the previous one, with a new mass of $m = 103.733 \text{ g}$. In this result, there are two problems: first, the beams are composed of multiple surfaces with different curvatures, which make the thickness extremely variable; second, the presence of four beams with a very small thickness. Instead, Figure 1.7 shows the result of the static simulation. As one can expect, the maximum internal stresses increase to a value of 29.411 MPa . Due to the reasons stated above, the structure is manually polished, removing the four critical beams and replacing the other beams with manually created ones, preserving the optimal direction found with the optimization process.



Figure 1.8: Stiffness maximization result after the trimming

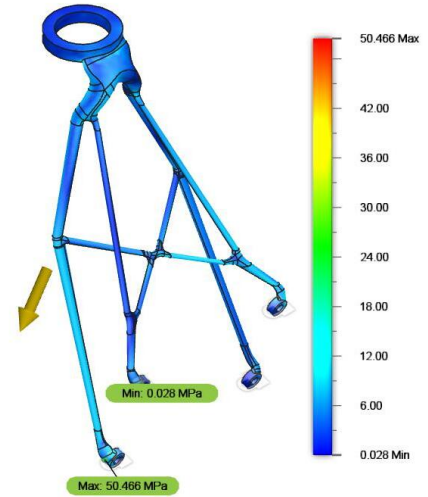


Figure 1.9: Static simulation after the trimming

In Figure 1.8 is reported the post-processed component, in which the mass is reduced to $m = 96.089 \text{ g}$. Instead, Figure 1.9 presents the static analysis. As it is possible to see from the static analysis, the component is still under critical condition, meaning that from a mechanical point of view, it may be possible to remove more material. Analyzing the structure it is seen that all the beams have a diameter in the range between 2 mm and 4 mm , and to decrease the instability problem of the structure it is decided to not minimize the thickness of these elements even further. Instead, the upper part is bigger with a thickness above 10 mm that could lead to thermal problems during the manufacturing process. Based on the arguments provided previously, it is decided to modify the upper part obtaining the result reported in Figure 1.10, which mass is $m = 69.824 \text{ g}$.



Figure 1.10: Component after reshaping the upper part

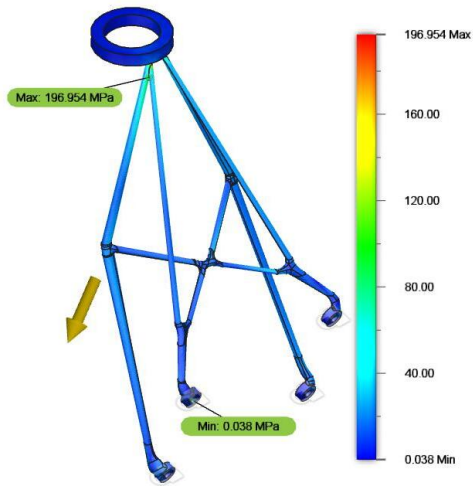


Figure 1.11: Static simulation after reshaping the upper part

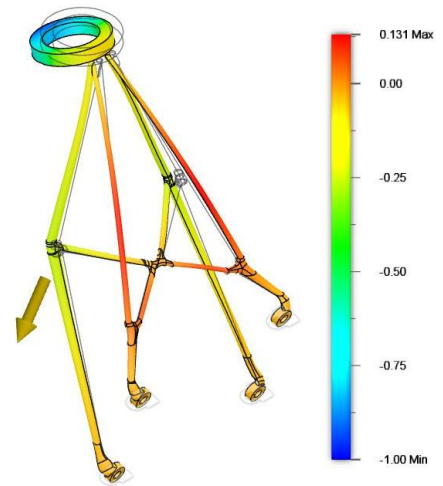


Figure 1.12: First mode shape after reshaping the upper part

In Figure 1.11 is reported the static analysis. It is possible to see that, even if the maximum internal stress increased to 196.954 MPa . Regarding the modal analysis, the first mode has a natural frequency of 72.529 Hz . Moreover, analyzing the participation factor it is observed that the predominant participation occurs along the z -direction, with a value of 54.61 , meeting the requirements. However, the participation along the x -direction, at 35.58 , though insufficient for it to be a dominant mode in that direction, is not insignificant. Examining the mode shape, particularly the component along the x -direction, as depicted in Figure 1.12 it is evident that the primary contribution arises from the rotation of the connector interface. Therefore, a final attempt is being made to relocate the connection between the beams and the connector interface to a more neutral position. This outcome is illustrated in Figure 1.13 with the final mass of this component $m = 73.597 \text{ g}$.



Figure 1.13: Final Part

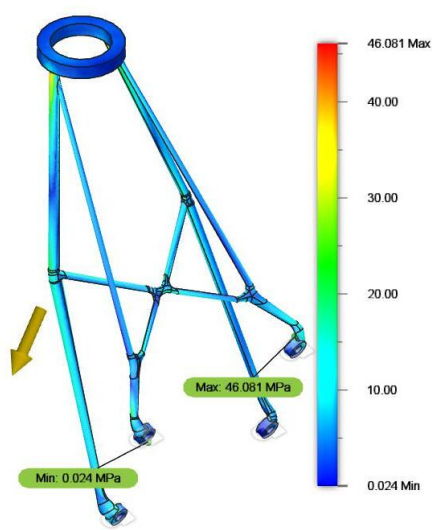


Figure 1.14: Static simulation for the final part

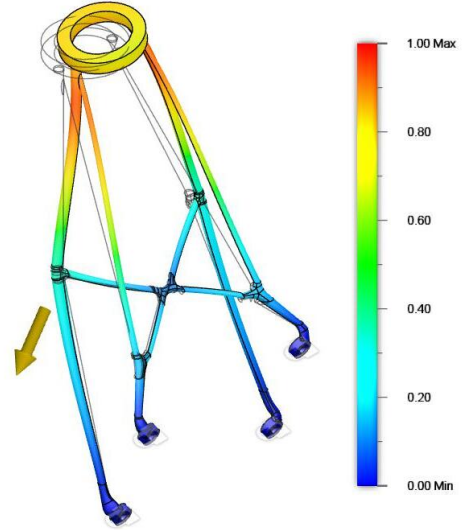


Figure 1.15: First mode shape for the final part

As depicted in Figure 1.14, with this change the critical part present in the previous static simulation disappeared, and the maximum internal stress decreased to 46.081 MPa . In Figure 1.15, instead, is reported the first modal shape linked to a natural frequency of 98.725 Hz . Analyzing the participation along the three directions, $\hat{x} = 0.0068$, $\hat{y} = 90.1175$, $\hat{z} = 0.0091$, it is observable to see how this mode is associated with the y – *direction* with a negligible contribution towards the other two. Therefore, it respects the frequency requirements.

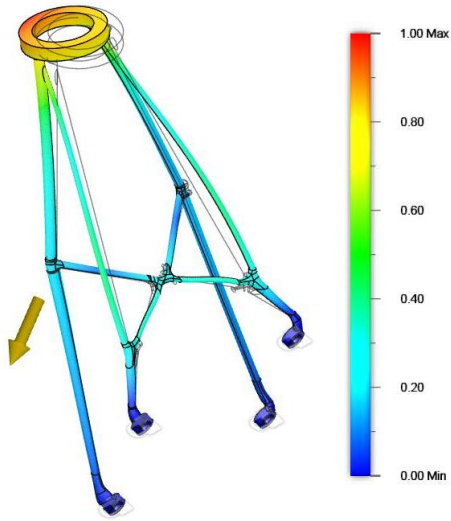


Figure 1.16: Second mode shape for the final part

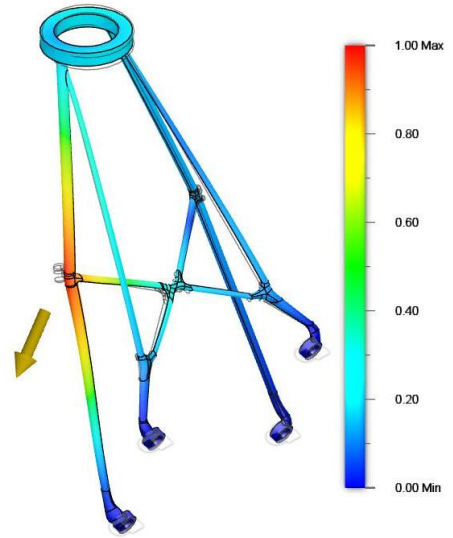


Figure 1.17: Buckling shape for the final part

Figure 1.16 reports the second mode shape associated with a natural frequency of $101.324Hz$. The participation factors are $\hat{x} = 7.0659$, $\hat{y} = 0.1035$ and $\hat{z} = 80.7709$. The participation factor shows how this is a mode along the z – *direction*, and independently from the fact that the contribution along the x – *direction* is negligible or not, the natural frequency is above the frequency requirement along the x – *direction*, making this solution compliant with all the requirements. Additionally, a buckling analysis is performed given the kind of structure obtained and the load condition. Indeed, the component is composed of beams that are compressive-loaded. This combination may lead to instability phenomena that could cause failure even with low loads. Figure 1.17 shows the buckling shape of the most critical mode. As it is possible to see, for this buckling mode the critical part is the beam on the left that connects the connector interface with the bottom bolt. For this mode, the safety factor is $SF = 14.567$, intended as the ratio between the load that could lead to buckling and the effective load. This means that buckling may happen with an acceleration load 14.567 times higher than the one predicted, or with a point mass constrained to the connector interface 14.567 times higher than the actual one. Therefore, the designed structure has a significant margin even at buckling.

Step - 2

Design for additive

Once the final optimization is obtained, the printing process analysis begins: in this first iteration, the component orientation is the primary variable improved. The main goal is the selection of an orientation to ensure good performance without focusing on the supports themselves. Concentrating on them at the beginning might lead to inaccurate and inferior overall properties of the component. Therefore, full support structures are used to prevent this issue.

To select the best possible orientation, Altair Inspire 2022.3 software is used, since it offers three different support options to choose from: minimize support deformation, minimize printing time, minimize supports. In the orientation that minimizes support deformation, the component is placed vertically, with the connector at the bottom, similar to the slightly tilted orientation that minimizes supports. On the other hand, in the orientation that minimizes printing time, the piece is positioned horizontally, parallel to the printing platform. For this option, the component is raised by 1 *mm* to prevent interference with the printing platform during the analysis and to avoid localized thermal deformation on the bolt insertions at the bottom. The supports adopted use the same material of the component, AlSi10Mg, with a thickness of 0.4 *mm*, which is double the value suggested by the software to reduce the time necessary to perform the analysis and avoid the creation of supports that might be too thin to be reproduced by the 3D printer. Since this is a qualitative analysis, the average thickness of each element is maintained at 5 *mm* to have a good preliminary estimation without high computational requirements. In this first iteration, the best orientation in terms of deformation and Von Mises stresses is chosen, without the contribution of the support type. The analysis results are listed below:

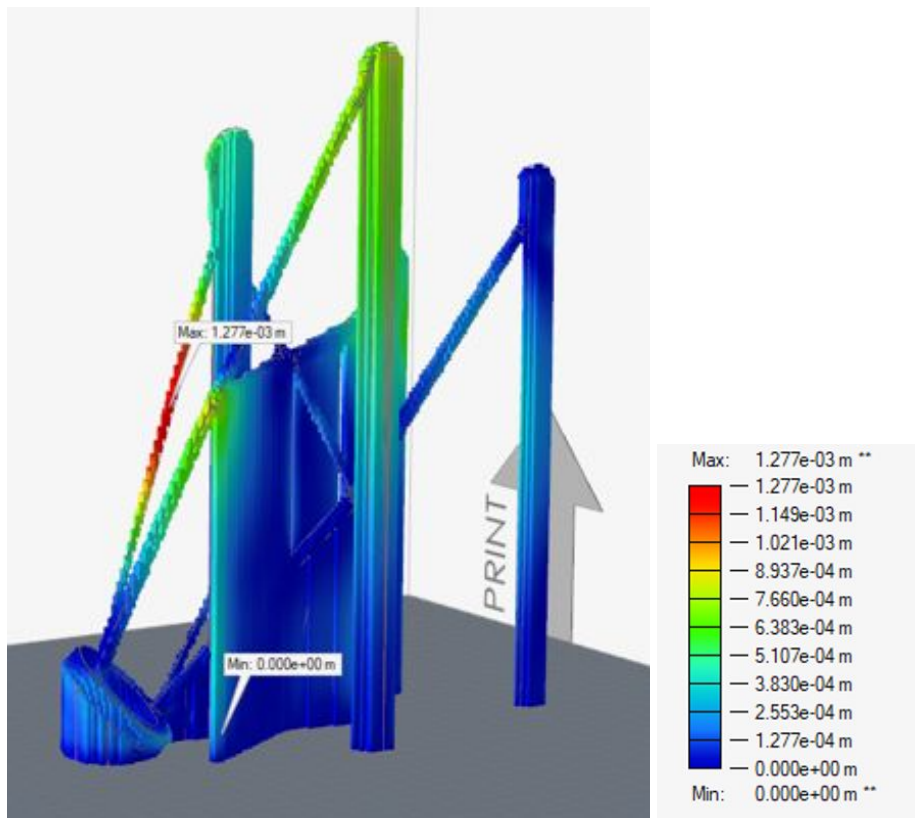


Figure 2.1: Orientation for minimization of supports deformation

The configuration with minimum support deformations in Figure 2.1 has the highest deformations on the component, with a peak of 1.277 *mm* around the mid-length of the left beam. As expected, on the supports the deformations are minimal, with more than one order of magnitude difference, compared to the other orientations. Despite this, the high amount of deformations on the component's beam is not negligible, so this solution is discarded.

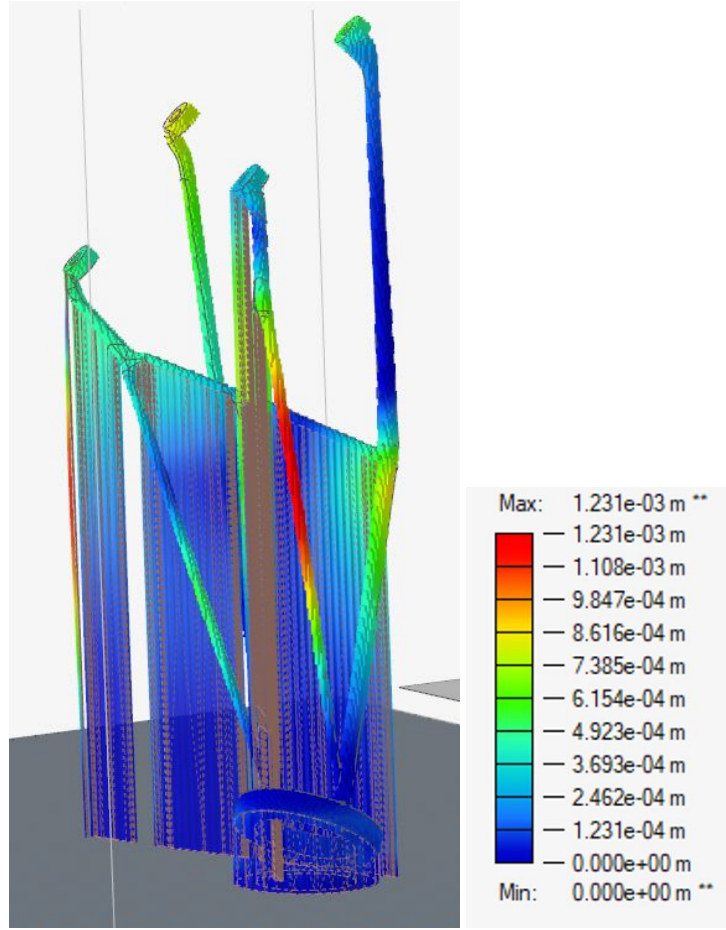


Figure 2.2: Orientation for minimization of supports

The analysis in Figure 2.2 is the worst overall, because it has displacements comparable to the first analysis, with a peak of 1.231 *mm*, but also deformations in the same order of magnitude on the supports. The supports, by deforming, cause the additional change of component shape during the printing as a consequence. For this reason, this configuration is discarded.

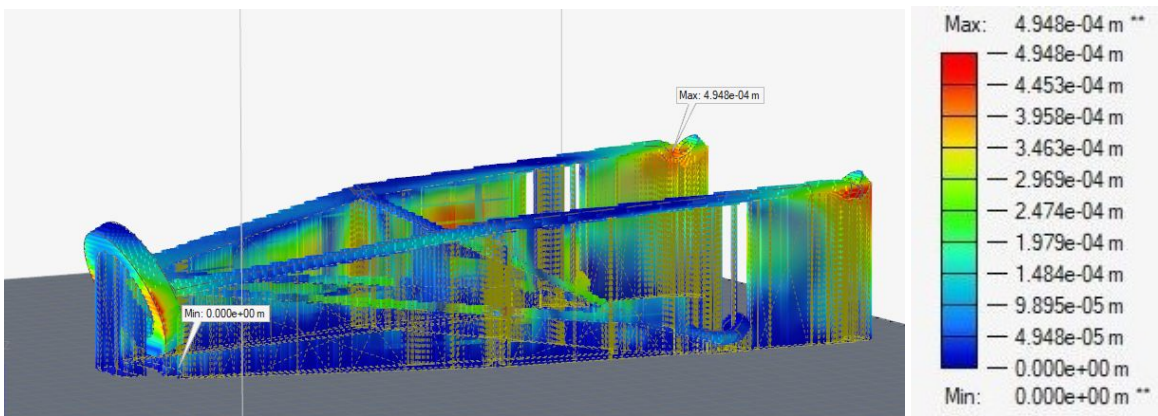


Figure 2.3: Orientation for minimization of printing time

The configuration shown in Figure 2.3 is analyzed with the minimum computational time because it involves fewer material layers. Additionally, it offers the best performance in terms of deformation, as indicated by its peak of 0.495 *mm*, which signifies deformation at a lower magnitude compared to the previous two cases. Additionally, the highest displacements are located on the top left bolt and connector sides. In contrast, the top left beam of the component does not get excessively deformed similar to the previous tests. For these reasons, it is finalized as the orientation choice.

Some similarities within the three configurations are that the support mass is maintained constant, but this can be explained by incorrect mass modeling of the full supports from the software: more detailed and correct data is evaluated in the following chapter. Also, the von Mises stress reaches the peak of 20.4 MPa which is the same for all three configurations, so supports do not influence this matter. Since the orientation is chosen, a secondary analysis is performed for the choice of support types: three main options are present, h-cells, rods, and tree supports. The first option creates hollow cube-like supports that get smaller as they get near to the component, the second creates vertical rods that maintain a fixed diameter, and the third creates a shaped branching structure that supports certain portions of the overhang, getting smaller as the layers go up. The aim is to obtain supports that reduce the deformation of the component while maintaining low support mass and a low value of springback [2] upon support removal. In this analysis, the software automatically generates supports, maintaining the previously used support thickness of 0.4 mm , but increasing the element thickness to 0.3 mm for more precise results. The printing characteristics mostly comply with the software standard, with minor modifications applied to each support methodology: for h-cell supports, the maximum subdivision level is set to 3 to ensure more precise supports as layers are added; for rod supports, the pin separation is doubled to 4 mm to reduce the proximity of the rods and increase the heat dissipation, while also doubling the rod diameter to 2 mm to prevent component deformation due to the increased distance between pins; for tree supports, the branching angle is set to 55° to avoid operating at the critical overhang condition of 45° .

timemin_hcellsupports	0.288 kg
MaxOptimizedv3 v11 ins...	0.074 kg
Support 1	0.201 kg
Support 2	0.004 kg
Support 3	0.009 kg

Total mass of h-cell supports case

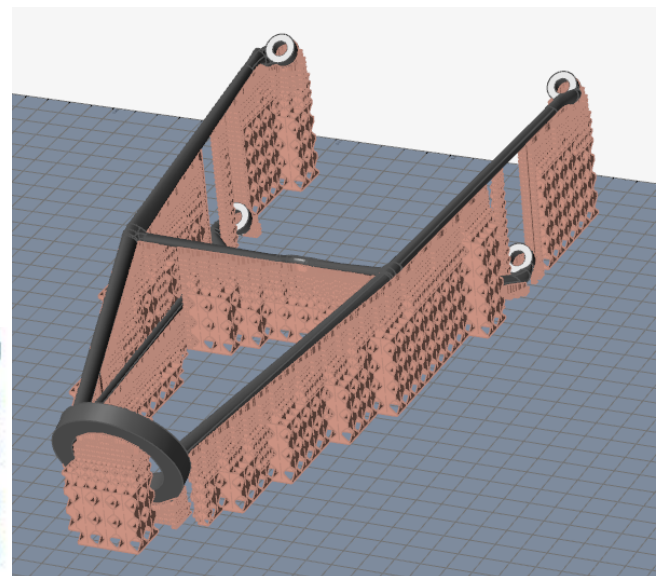


Figure 2.4: Supports in h-cell configuration

timemin_rodsupports	0.093 kg
MaxOptimizedv3 v11 ins...	0.074 kg
Support 1	0.018 kg
Support 2	$3.144 \times 10^{-4} \text{ kg}$
Support 3	0.001 kg

Total mass of rod supports case

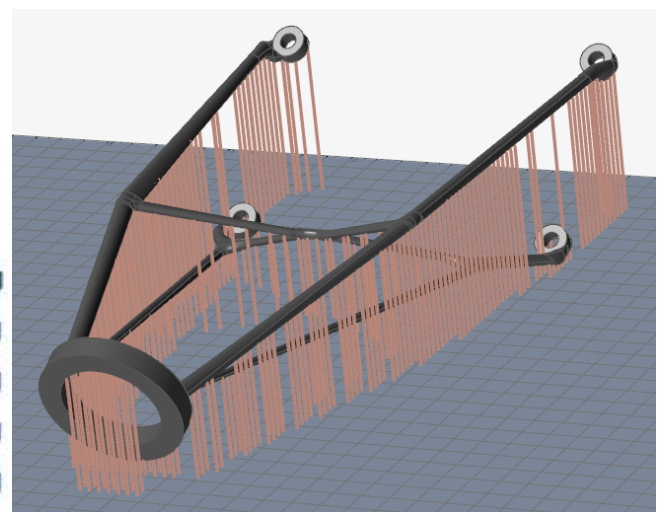


Figure 2.5: Supports in rod configuration

timemin_treesupports	0.167 kg
MaxOptimizedv3 v11 ins...	0.074 kg
Support 1	0.086 kg
Support 2	0.003 kg
Support 3	0.004 kg

Total mass of tree supports and component

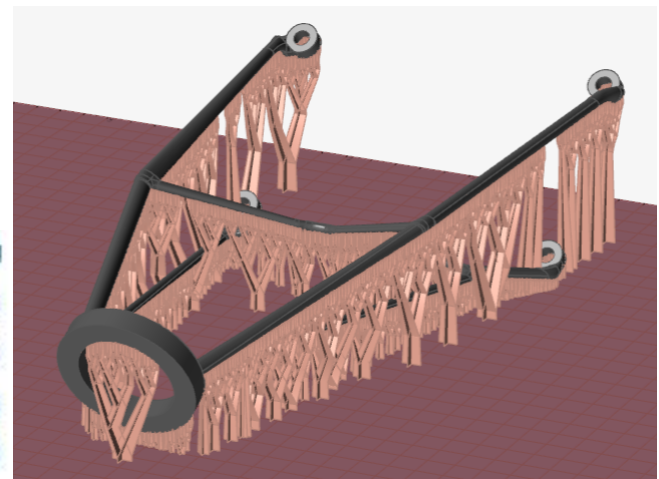


Figure 2.6: Supports in tree configuration

The h-cell supports of Figure 2.4 are the heaviest because they occupy a bigger base than the other two, but this can favor the minimization of deformation by providing a more robust surface for supports. The rod supports of Figure 2.5 are the lightest and thinnest due to the modification explained before, in this way, they do not create invasivity on the component because their structure makes it easy to remove. The tree supports of Figure 2.6 are in the middle in terms of mass, but they do not maintain symmetry on the component because of the uneven support branching, leading to an unsymmetrical distribution of deformation.

It is observed that each support methodology has 3 subdivisions throughout the component: below the beams and the four bolts (support 1), below the connectors (support 2), and inside the connector (support 3). As one can easily see, support 1 is the heaviest region, because it holds the majority of component overhangs. Since the supports related to the connector are not a critical region, no mixed support structures are considered. The final support choice is discussed in the next section since their selection is simulation-driven in terms of post-processing displacements and springback analysis.

Step - 3

Simulation

3.1 Supports Configurations

At this stage, the support types must be evaluated to select the structure with favorable characteristics in terms of mass, deformation, intrusivity and springback. At first, the displacements are compared, followed by the evaluation of springback resulting from support removal, to identify which structures pose the highest risks of permanent deformations.

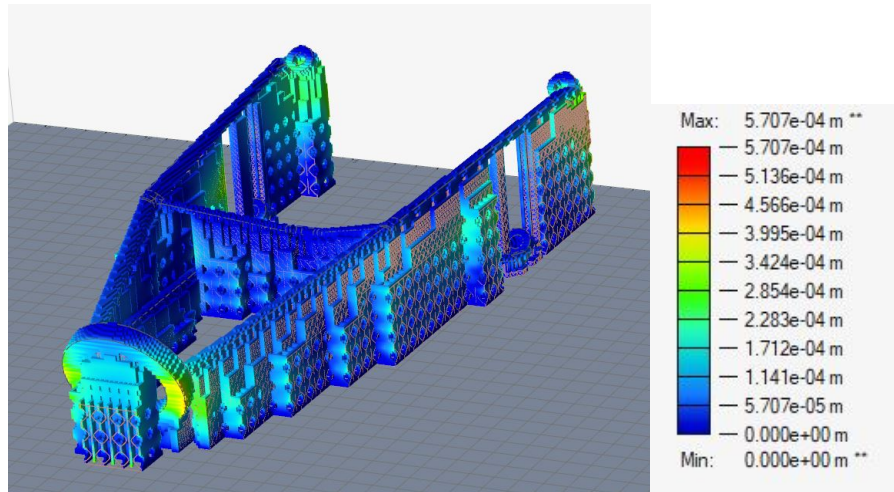


Figure 3.1: Displacement field distribution on h-cell supported component

The configuration with h-cell supports in Figure 3.1 exhibits the lowest displacement on the supports, with the most critical conditions observed below the top beams above the connector's sides, where the supports are absent, and on the supports connected with the top bolt insertions. Medium-to-high deformations are observed on the connector sides, mostly on the parts furthest from the supports.

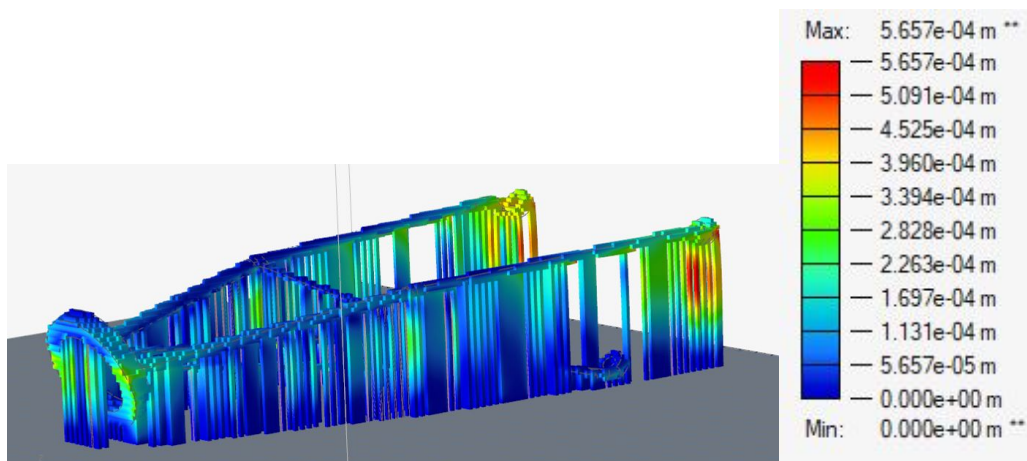


Figure 3.2: Displacement field distribution on rod supported component

The setup with rod supports in Figure 3.2 presents the lowest displacements on the components, having a quite similar output to the h-cell support structure, but with only half the support mass. The critical areas are the very base of the top bolts insertions and the support area below them.

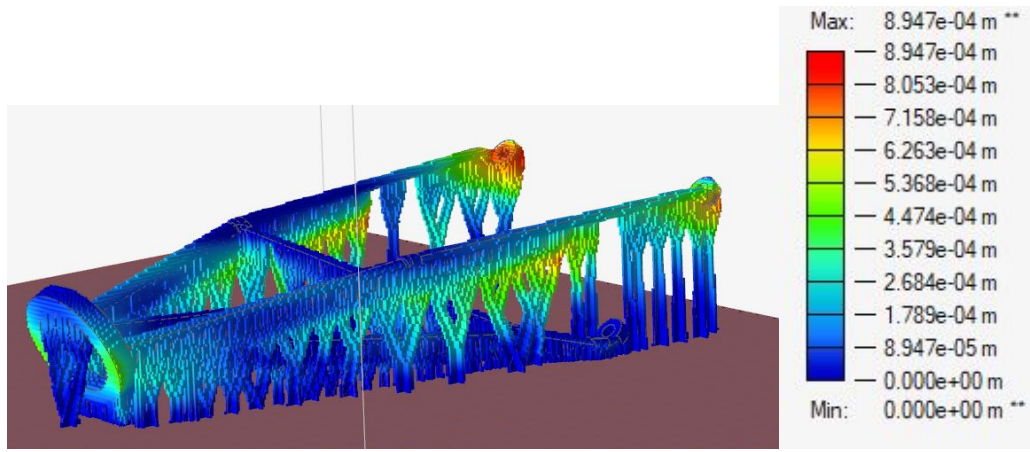


Figure 3.3: Displacement field distribution on tree supported component

In the tree support structure shown in Figure 3.3 the highest deformations are observed, extending the critical region through the full top left bolt insertion. In addition, these supports create uneven stresses because of their unsymmetric branching, leading to having only one bolt with the highest deformation, making the final piece unbalanced. From the three analyses, the chosen support architecture is the rod support option due to the smallest mass present and good deformation performance.

At this point, the deformation magnitude of the post-printing component is evaluated in terms of springback. This phenomenon occurs when the supports are removed, which may cause the component to bend and retain plastic deformations.

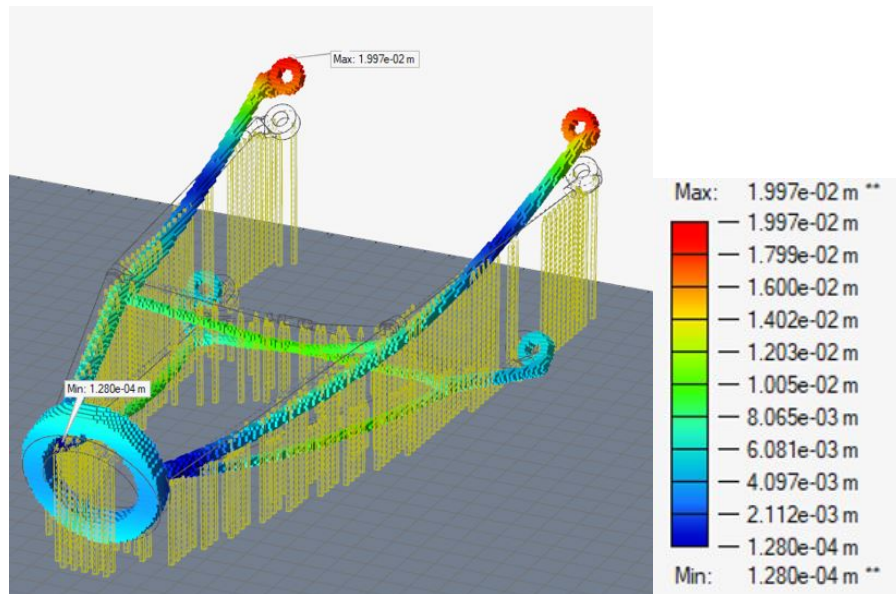


Figure 3.4: Springback effect on the component

Figure 3.4 shows the highest springback at the terminal top insertions, causing high elastic displacements on both beams, while the connector has the smallest effect because that part is very near to the printing bed and so it is less affected. It is worth noting that this effect is not dependent on the mass of the geometry but only on their position and type, differently from the displacement.

3.2 Compensated Geometry

The first iteration provided a component with residual deformation caused by the printing process. A solution to remove the plastic defect is to conduct a second iteration using a compensated geometry that preventively

exploits the bending effect to remove the plastic defect. This compensated geometry induces a displacement on the original component such that, when the supports are removed, the printing-induced deformation and the deformation added by the compensation are recovered by springback, which makes the final element equal to the software geometry, without residual deformations. At this stage, the compensated geometry can be extracted and analyzed as a second step. This geometry becomes the definitive printed piece.

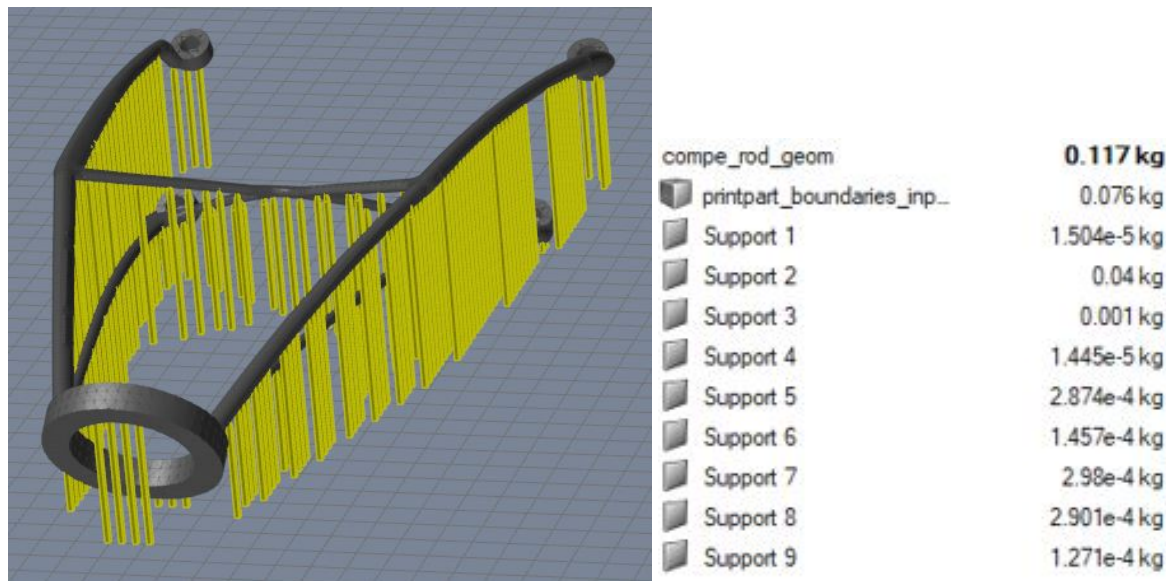


Figure 3.5: Supports distribution on the compensated geometry

Figure 3.5 displays the new component bent towards the printing bed so that the bending caused by the spring-back pushes the piece upwards. This time the support regions are nine: four below the top bolt insertions, two below the inner bolt insertions, two inside and below the connector and one for the entirety of the beams. This guarantees a more controlled springback since it must be able to bend the object in the desired configuration. The different distribution of the supports varies their mass, making them slightly heavier.

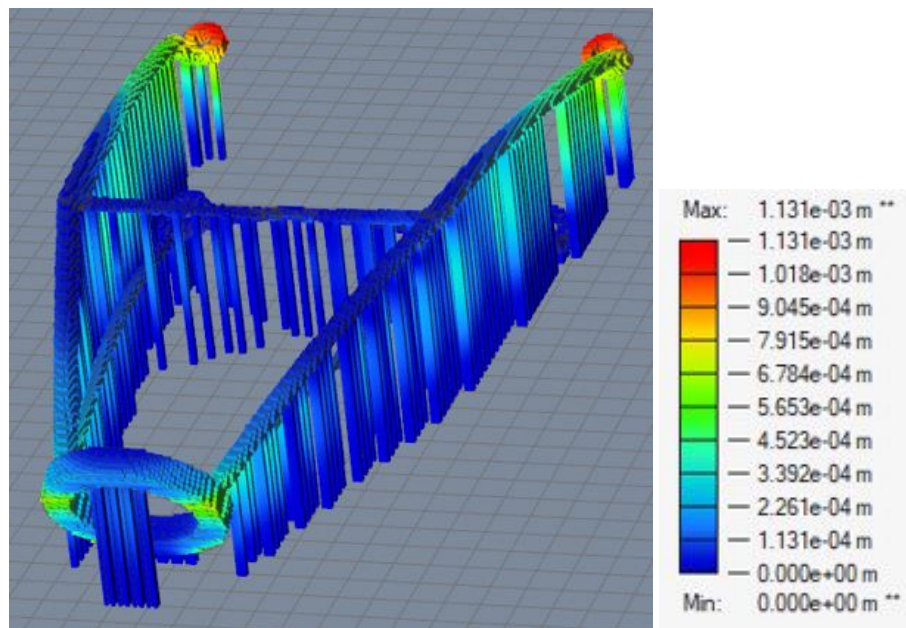


Figure 3.6: Deformations on the compensated geometry

Figure 3.6 shows the new configuration with the deformation peak twice as high as the initial component's. This is because the compensated piece must be subjected to the deformation to springback and reach its desired shape. Additionally, the peak displacement is located on top of the upper bolt insertions, differently from the

old geometry, Figure 3.2, where it is located on the bottom, because it is the section that must be deformed the most towards the top.

3.2.1 Springback

Upon removing the supports, the component recovers its predicted geometry. Figure 3.7 shows the final shape compared with the printed one and the displacement from the latter. The highest displacement occurs, as expected, at the ends of the legs of the support, since those are *free ends*.

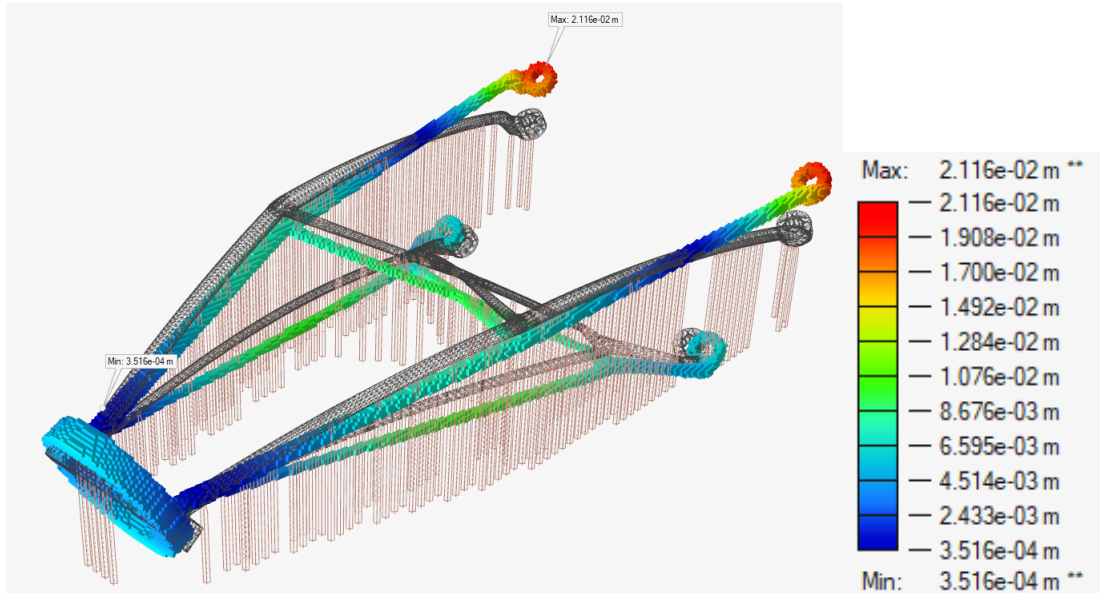


Figure 3.7: Recovered geometry after the support removal. The color scale shows the displacement from the printing geometry

Comparing the springback effect of Figure 3.7 to the non-compensated geometry case of Figure 3.4, there is a displacement disparity as expected. Using the peak values of displacement as an example, it demonstrates that the difference between the peak springback in Figure 3.7 and Figure 3.4 equals exactly the sum of the peaks in Figure 3.6 and Figure 3.2, providing the efficacy of the compensated geometry. The von Mises analysis of Figure 3.8 shows stresses that are very high due to the aggressive topology optimization adopted. However, although their peak of $204MPa$ is approaching the yield stress of $220MPa$ reported in [Tab. 3], it is still below the limit, thus they do not cause plastic defects in the printed component. Moreover, adopting a compensated geometry does not increase the peak stress in the object. The section with the highest residual stresses is the connector because it is a part of the geometry that must not be deformed and so it retains the most stresses.

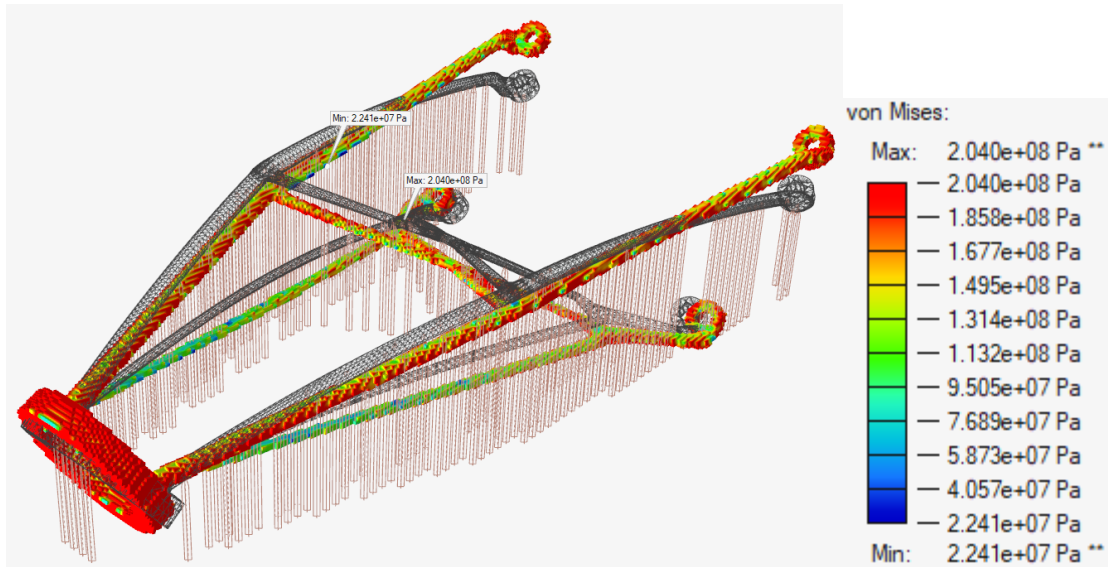


Figure 3.8: Von Mises stresses after the support removal and springback

3.2.2 Print Analysis

After producing the compensated geometry, a printing analysis is performed obtaining a printing time of 63 281 s and a support volume of $1.284e - 3 m^3$. Figure 3.9 shows that the collision distance remains positive throughout the manufacturing process, indicating no interaction between the print and the recoater.[3].

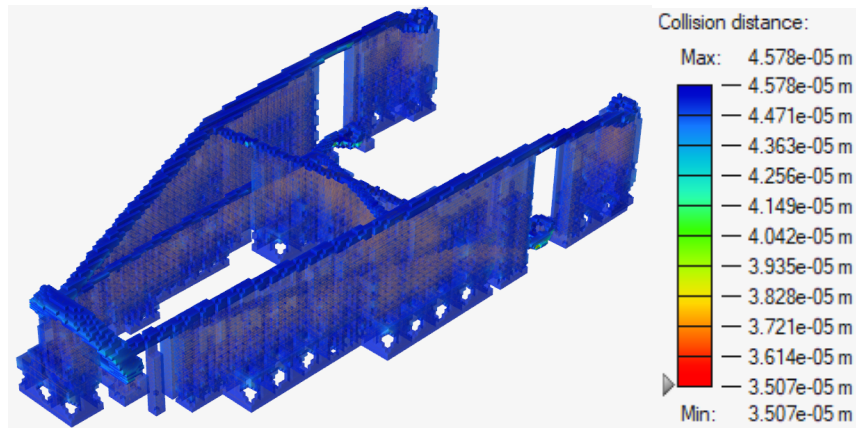


Figure 3.9: Collision evaluation on the compensated geometry

Additionally, a layer shifting analysis is performed to evaluate the surface roughness of the component. Once a layer is printed, it cools down producing a certain shrinkage, while the subsequent layer is printed according to the model's dimensions. This phenomenon leads to a shift between the layers and thus an increased roughness, as shown in figure 3.10. The results indicate a layer shift ranging from $0 \mu m$ to $4.274 \mu m$, which, in the worst area, corresponds to a *local roughness* of N9 ISO grade [4]. Given the size of and the use of the item, this amount of layer shifting does not represent an issue.

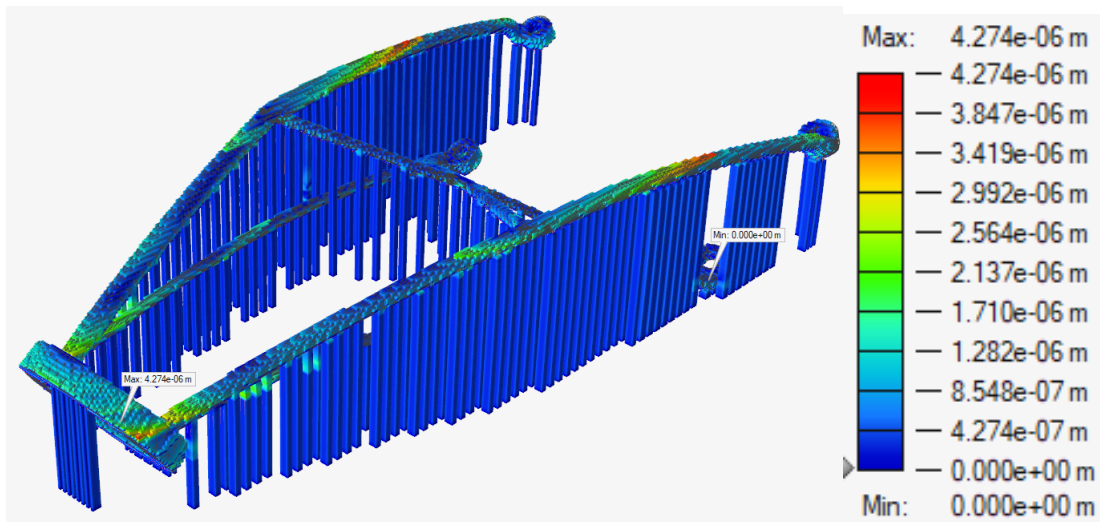


Figure 3.10: Layer displacement during printing on the compensated geometry

Chapter - 4

Final performances and conclusion

The topology optimization method proves efficient in reducing the component mass while maintaining good performance. Considering the performance requirements for the natural frequency and acceleration, many iterations are generated, resulting in a low mass structure with one-sixth of the original assembly mass of 453g. This optimization maintains high margins in terms of natural frequencies in all three directions and achieves a QSF of 1.25, as specified in the Vega User Manual [1].

The additive optimization provides a horizontally oriented structure with minimal deformation, using the lightest and least voluminous support option—rod supports—among the three available types.

The printing analysis highlights the presence of residual displacement in the component. which is resolved through the implementation of compensated geometry. This adjustment completely eliminates the residual stresses and leads to a more evenly distributed region of high stresses, particularly in the connector, while keeping all stresses below the yield threshold, with the added downside of raising the overall masses of supports and component by a few grams.

Design for performance		Design for additive		Simulation	
Mass [g]	73.597	Mass [g]	75.552	Printing Phase	
Maximum Stress [MPa]	46.081	Support Mass [g]	41.39	Maximum Residual Stress [MPa]	204
Maximum Displacement [mm]	0.441	Support Volume [cm^3]	15.502	Maximum Displacement [mm]	1.131
Maximum Deformation [-]	0.001	Orientation [-]	time minimum	Printing Time [s]	63 281
x Natural Frequency [Hz]	245.545	Support type [-]	rods	Springback	
y Natural Frequency [Hz]	98.725			Maximum Residual Stress [MPa]	204
z Natural Frequency [Hz]	101.324			Maximum Displacement [mm]	21.16
Buckling safety factor [-]	14.56			Support volume [dm^3]	1.284

Table 4.1: Final Results

Bibliography

- [1] Arianspace. Vega user's manual, issue 4 - revision 0. Technical report, Arianspace, April 2014.
- [2] Steve Benson. Springback definition, [Link](#).
- [3] Altair Engineering. Result types for slm, [Link](#).
- [4] International Organization for Standardization. Iso - geometrical product specifications (gps) - indication of surface texture in technical product documentation, 2002. Amendment 1:2008.